

Problem #3: Bow or Onion (15 marks)

Solution:

Depending on the sense of the word **luk**, different sets of words co-occur with it in the paragraphs. For example, when a paragraph is talking about **luk** = *onion*, the following words are more likely to co-occur:

ovoschi (=vegetable), *edjat* (=eaten), *listja* (=leaves), *rastenie* (=plant), *chesnok* (=garlic)

Whereas when **luk** means *bow* in a paragraph, the following words are found to co-occur more often:

strely (=arrow), *okhotnikov* (=hunter), *oruzie* (=weapon), *metanija* (=shoot), *ubit* (=kill),
strelbe iz luka/ strely iz lukov/ strelba iz luka (archery)

By looking at the common words in the reference paragraphs and the test paragraphs, one can guess the sense of the word **luk** for each of the paragraphs. Note that there are other words like *chem* (=than), *i* (=and), *ego* (=its), which can occur with both the sense of the word **luk** (in fact, these can occur with any word), and hence are not useful in discriminating between the different senses of a word.

In paragraphs 2, 4, 6, 7, 10, **luk** = *onion*.

In paragraphs 1, 3, 5, 8, 9, **luk** = *bow*.